

Chapter 178

Mix Automation

The Fairlight page provides straightforward, flexible, and powerful automation capabilities for audio mixing.

Automation data can be created graphically by drawing automation changes, or by recording on-screen mixer or effects controls movement and changes, using a mouse or Fairlight control surface. You can even create panning automation based on the movements of on-screen people or objects in the Fairlight Viewer.

Once recorded, mix parameter automation for any control type—faders or sends, panning, or effects—can be played back in perfect sync with the audio and video.

This chapter covers automation mixing using the combined controls of the on-screen mixer and timeline.

For more information on basic mixing operations, see *Chapter 177, “Mixing in the Fairlight Page.”*

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Creating Mix Automation

The Fairlight page includes a comprehensive toolset for creating and editing mix automation. You can draw automation data, or record mix changes by moving on-screen controls, or you can use IntelliTrack-powered Audio Panning to Video, which writes panning automation based on the movements of people or objects in the Fairlight Viewer. Yes, this really is “a thing.”

Why Use Track Automation?

If you’re a video editor, you may be more familiar with using clip-based automation to make changes to your audio mix, but this approach has limitations. Sometimes there are global changes you may want to make to a track over the timeline or want to adjust groups of tracks at once see *Chapter 179*, “Track Groups.”

When working with time-based effects like reverb or a repeating delay, you may want to dynamically adjust the level of “trails” as the track evolves. For example, you may have a scene that changes perspective from a larger space to a smaller one, and the room ambience or tonal quality of the dialogue needs to change over time as the characters move through the scene.

This is where track automation techniques (whether drawing or recording the movement of controls) can be powerful and creatively inspiring.

What Can Be Automated

Automation can be recorded for nearly every control in a Mixer channel strip, including the fader, sends, mute, and panning. You can also automate audio plugin parameters as well.

Audio Panning to Video

IntelliTrack-powered Audio Panning to Video accelerates creativity by automatically tracking people or objects in the Fairlight viewer to generate pan automation that matches their on-screen movements with the precision afforded by Resolve's state-of-the-art object tracking engine.

Drawing Automation

If you're a video editor or audio creative, drawing mix automation data may be more straightforward in many contexts, when making static or ramped level changes or switched changes to controls such as a mute button, especially if you are using a mouse without the benefits of using Fairlight hardware control.

For example, trying to finesse a real-time "performance" of fader movements or effect parameter changes, you could select a timeline range and mute several tracks at once, "carve out" fixed level boosts or drops on tracks at the exact point you want them to occur based on an on-screen event or dialogue, or slowly increase the strength of a reverb applied to Foley footsteps as a character walks into a long, dark cavern.

When drawing automation, you don't need to pay attention to the various Fairlight automation recording modes or control enables/disables for the automation parameter you want to create or edit; you can just draw or adjust your automation data in the corresponding track "lane" on the Timeline.

Recording Automation

You can also create mix automation by recording real-time on-screen fader, control movements, or plugin parameter changes during playback. Afterwards, your "performance" can be accurately played back. Another advantage of recording your automation is that you can intuitively "perform" the changes against the picture as actions occur.

In this way, you can create a dynamic mix where different audio levels, pan, EQ, dynamics, and audio processing settings change over time to fade music up or down against effects or dialogue, pan the sound effect of a car driving by from one speaker to another, or gradually increase the strength of a reverb effect as your character walks into that long, dark cavern.

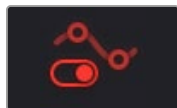
While the recording of keyframe automation is commonly associated with either the on-screen mixer or the Fairlight console, you can also record automation using controls found in the Inspector, or using the controls of the Pan, EQ, and Dynamics plugin effects, thereby enabling you to record automation for the various audio effects that you've applied to a track.

Drawing Mix Automation

Automation can be drawn in the Timeline for any parameter by globally enabling automation recording and playback, viewing a parameter to be automated, then drawing the automation in the Timeline.

Enabling and Viewing Automation

The Toggle Automation button, to the right of the transport controls, turns automation recording and playback on and off. It must be enabled to draw or record track-based automation and to see busses on the Timeline.



Toggle
Automation button

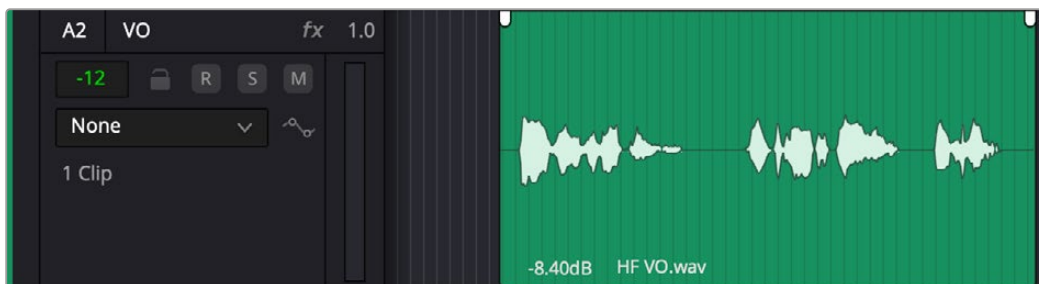
TIP: You can quickly enable/disable automation safe on or off for a selected group of channel strips, or to all channel strips, by holding down the Option key (Mac) or the Alt key (Windows) for all selected tracks or Command-Option (Mac) or Control-Alt (Windows) for all mixer channel strips prior to performing the operation. These shortcuts can save a lot of time in your workflow.

Automation View

Audio tracks normally display an arrangement of clips on the Timeline (“Clips View”).

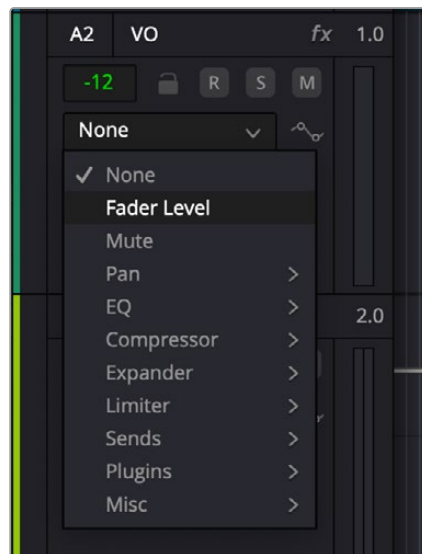
In Clips View, clips can be placed or edited, and clip gain for individual clips can be adjusted, see *Chapter 176, “The Fairlight Inspector.”* Effects can also be added to clips, which can be managed in the Inspector.

Tracks can also show an Automation View, which appears when the automation button is enabled. When active, a dropdown menu appears under the track controls in the track header. By default, the menu will display “None,” indicating you are working with clips only; no automation data will appear.



Fairlight Automation View showing “None”

Clicking on the dropdown menu allows you to choose an automation parameter to view or edit. Choices are included for all parameters that can be controlled on a track, including fader, sends, pan, mute, and built-in dynamics and EQ as well as AU or VST plugin effects.



Fairlight Automation View menu choices

For a faster workflow, you can right-click the Mixer channel control or parameter on a Fairlight FX plugin, supported Audio Unit, or VST plugin you want to view or edit.

Once a parameter is chosen, the automation view changes so that the focus is on editing track automation data rather than manipulating clips. The clips darken and the automation data line is shown in light gray. When automation is drawn or recorded for the parameter, the line will change to bright green.

NOTE: When in Automation View you can still move and trim clips, but cut/copy/paste and delete operate on automation keyframe data only. When in Automation View, timeline-based clip gain keyframe adjustment is not available, but overall clip gain can still be adjusted using Volume slider in the Audio tab of the Inspector.

Vector-based Automation

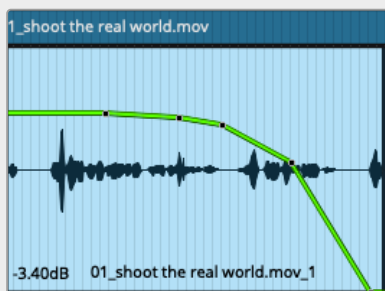
Audio keyframe automation is vector-based, so levels will smoothly ramp between any 2 points on the Timeline based on the direction of the change in control value and the time between the points.

For example, if you have a single keyframe setting fader level to 0 dB that occurs at 01:00:20:00 on the timeline, and then add a point at 01:00:28:00 at -10.5 dB, the level will change smoothly between 0 dB and -10 dB over the 8 seconds between the 2 points.

This allows you to draw individual keyframes to quickly construct different transition shapes, such as fade-ins or fade-outs, with just a few points, or adjust them up or down by specific amounts quickly and easily.

Creating Curves

Keep in mind that audio automation keyframes are not Bezier keyframes, so they only let you create a single level adjustment at that point in time. If you make a large adjustment with only a single new keyframe, you can end up with an abrupt level change, or level changes that don't have a curve you may want (exponential changes can be particularly useful on fade ins or outs to get a natural result). You can create convex or concave shapes when drawing by adding just a few keyframes to create your curve, or you can perform the change exactly as you want by recording the automation (see "Recording Mix Automation" later in this chapter).



A fade out transition with 5 keyframes for an exponential shape

Track Automation Line

Audio automation on a track is represented by a line that extends over length of the Timeline, called the automation line, or "curve." The level of the line can be changed by dragging, entering keyframes one at a time, or using the Pencil tool to draw a series of keyframes.



Automation line for fader level showing multiple keyframes

Sample Accuracy: Audio keyframes allow creation of precisely placed parameter changes on timeline tracks. On the Fairlight page, audio keyframes are sample accurate, meaning they can be placed at any sample location with sub-millisecond precision, allowing more precision than keyframe handling on the Cut and Edit pages. However, sample-accurate audio automation data created on the Fairlight page can be accurately played back on the other pages.

Sample accuracy is important when working to picture as the location of audio events often does not fall on exact frame or sub-frame boundaries.

Working with Audio Keyframes on the Timeline

To draw automated changes to any audio control:

- Enable the Automation button to see Automation View on all audio tracks.
- On the track(s) where you want to create automation, choose the parameter you wish to automate in the Automation type dropdown menu in the track header.
- Using Focus Mode's Multi tool allows you to draw automation using the up/down arrow cursor (to adjust an overall curve), enter or delete keyframes one a time using modifier keys, create different curves depending on individual keyframe position, select and delete a range of keyframes, and more.

Adding and Deleting Individual Keyframes

Managing keyframes on the automation line is nearly identical to working with clip gain.

- **To add a keyframe:** Hold down the option key and a plus sign appears. Click to place a keyframe on the automation line.
- **To remove a keyframe:** Hold down Command-option and a minus sign appears. Click on keyframe to delete it.

For more information see *Chapter 176, "The Fairlight Inspector."*

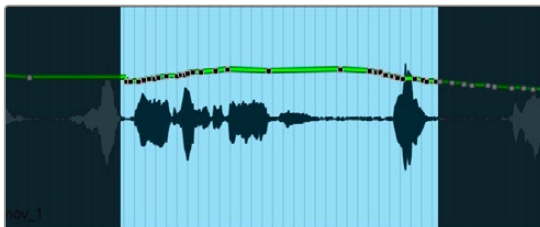
TIP: You can quickly flip between Clips View and Automation View using the keyboard shortcut F4. This can be changed by using the DaVinci Resolve > Keyboard Customization dialog.

Editing Mix Automation Curves

Using Edit Operations

Normal editing operations such as cut, copy, paste, delete, and nudge can all be used on automation keyframe data. When automation data is cut or deleted, the value at either end of the selection is preserved.

Nudging can be very useful when fine-tuning the location of automation events.

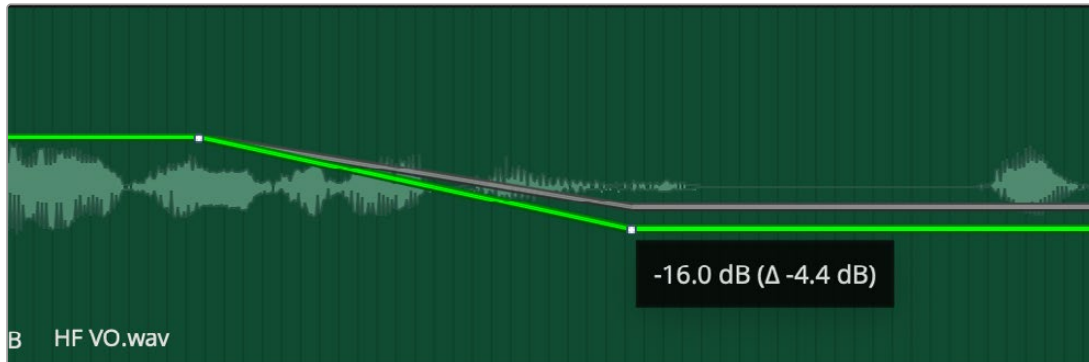


Before and after cut or deletion

Adjusting the Automation Line

Once one or more keyframes are on the automation line, the resulting curve for the entire track can be adjusted. Individual keyframes can be edited up or down using the same techniques as with adjusting clip gain; the whole line can be adjusted, or a selection can be affected.

If you adjust the last keyframe on a line with multiple keyframe points, the line will change from that point onwards to the newly set level.

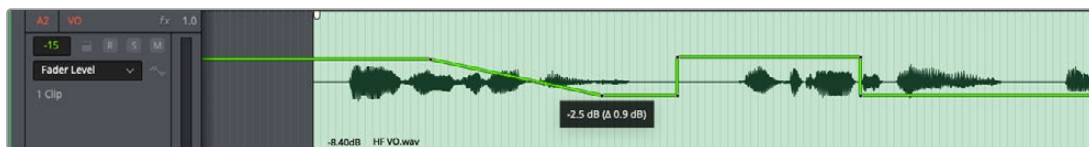


Adjusting an individual keyframe changes the line from that point onwards.

Adjusting the Entire Line

To adjust the entire line:

- With automation view enabled, click anywhere within the track, and select all automation data on the curve by choosing Command-A.
- Position the cursor close to the line and the dual arrow cursor appears, allowing the level of the entire line to be adjusted.



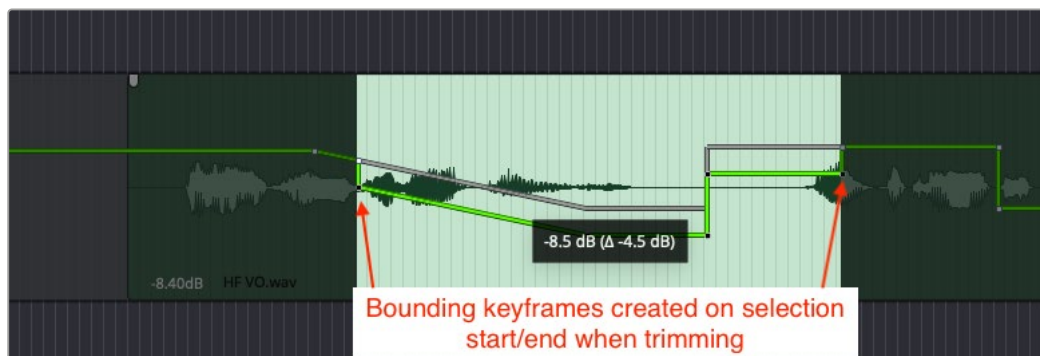
Select all automation to adjust the entire line

Adjusting a Selected Range

You can perform adjustments to only a specific range of the automation curve. This can be useful when you want to make “snapshot” changes for a specific scene.

To adjust only a selected range:

- Use the Focus Mode Select tool (I-beam cursor) to select the range you want to affect.
- Position the mouse close to any area of the line, and the up/down arrow cursor appears.
- Trim the line up or down. A tooltip will appear showing the current value and a delta to the original value, so any difference in a control’s value can be seen.
- At the bounds of the selection, “bounding keyframes” are automatically created, so the level at the edge of the selection is preserved.



Trimming a selected range of the automation curve

TIP: Hold the Shift key when adjusting levels to maintain precision. When adjusting fader levels this allows .1 dB accuracy, even when working with small track heights.

Drawing Automation with the Pencil Tool

The Pencil tool allows automation keyframes to be drawn freely:

- Choose the Pencil tool in the Automation toolbar, or when in Focus mode, hold down Option-Shift to temporarily switch to the Pencil tool.
- Click the mouse and drag to draw keyframes in either direction.

Automating Switched Controls

Some automation parameters are switched (i.e., they have only fixed states). Examples of this type of control are mutes (on or off). When adjusting switched automation, keyframes can only be at the bottom end of the Automation view or the top.



Mute automation showing on and off states

Trimming Automation from Unity in the Mixer

Trim mode supports automation trimming from unity (Fader set to 0) in the Fairlight mixer by following the steps below:

- 1 On the Automation toolbar, set the Automation mode to Trim.



The Automation toolbar - Trim mode

- 2 Choose a Touch mode (Latch, Snap, or Snap Latch).

- 3 Click the Automation Arm button on the mixer channel or tracks you want to trim. The faders on the armed mixer channels will turn yellow and snap to zero, and the dB Indicator above each armed fader will read $\Delta 0.0\text{dB}$.

TIP: You can either Command-Option click the Automation Arm button to trim automation on all tracks or Option-click the button to trim automation for a collection of tracks or mixer channels you've selected.



Mixer faders set to trim automation from unity

- 4 Begin playback and move the fader above or below $\Delta 0.0\text{dB}$. The automation curve will be trimmed, and the dB indicator will display the amount of trim as a plus or minus (+/-) value.

If you've chosen Snap mode, the fader will return to $\Delta 0.0\text{dB}$ based on the Glide time set in Preference > User > Fairlight > Automation.

Audio Panning to Video

Audio Panning to Video uses DaVinci Resolve's state-of-the-art object-tracking engine to help you quickly generate precision panning automation based on the movements of people and objects in the Fairlight Viewer.

Audio Panning to Video offers Automatic and Manual modes of operation.

Automatic Tracking

In Automatic Tracking mode, you can select a person or object on screen, and DaVinci Resolve will automatically process the related movement data across a selected timeline range, and then generate the automation data.

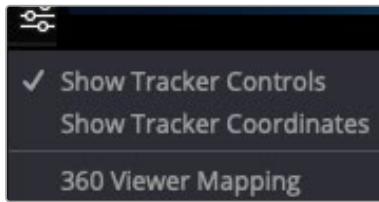
To enable and use Automatic Tracking:

- 1 Click the Toggle Automation button to the right of the transport controls.



Toggle Automation button

- 2 Click the Fairlight Viewer tools menu and choose Show Tracker Controls.



Fairlight Viewer Tools menu

The Tracker Controls will appear along the lower edge of the Viewer, and you'll see an Auto Tracker Object on screen.



Tracker controls



Audio Tracker Object

You can also select Show Tracker Coordinates, which provides an alpha-numerical representation of the tracker's current on-screen position.



Audio Tracker Object
with Coordinates

- 3 Select the target track you want to write panning automation to, and use the I-Beam tool to select the Timeline range you want the tracking to work on.



Range Select tool

NOTE: Auto-tracking will not occur unless you select a Timeline range.

- 4 Click the Auto button in the Tracking Controls section to activate Auto Tracking.
- 5 Based on the panning movement you want to automate (horizontal or vertical), click the Left-Right (L-R) or Down-Up (D-U) button.

TIP: The Front-Back (F-B) button is only available when the Auto button is off.

- 6 On the target track, click the Automation dropdown in the Track Header, then followed by Pan > L/R Pan or Pan > U/D Pan to match your selection in the previous step.
- 7 Move the Auto Tracker Object on top of the on-screen object or person you want to track.
- 8 Click the Track button to begin the Auto-Tracking process.
- 9 Start playback to check your audio track. It should follow the object or person you selected for tracking.

TIP: If the track is more complex, for example, a person who is changing their position relative to the camera or may have moved out of frame, you may need to do your tracking in sections.

In this instance, you can select an area that wasn't tracking well, move the tracker object into position, and track over the problem range to redo the track for that portion.

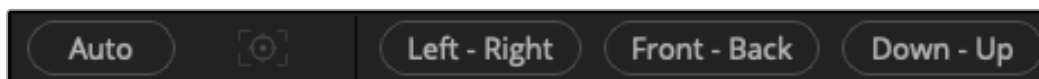
Manual Tracking

Manual mode lets you create keyframes based on the Playhead position on the Timeline, and DaVinci Resolve will smoothly vector between the keyframes like any other automation curve data.

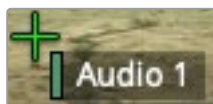
To enable and use Manual Tracking:

- 1 Click the Toggle Automation button to the right of the transport controls.
- 2 Select the target track that you want to write your panning automation to.
- 3 Click the Video Viewer tools menu and choose Show Tracker Controls.

The tracker controls will appear along the lower edge of the Viewer, and you'll see a Tracker Object on screen as a simple crosshair.



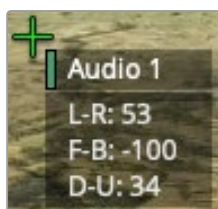
Tracker Controls manual mode



Manual Tracker Object

NOTE: The Tracker object will not appear onscreen unless an audio track has been selected.

You can also choose to Show Tracker Coordinates.



Manual Tracker Object with Coordinates

- 4 Ensure that the Auto button is inactive in the Tracking controls. Now you're in Manual Tracking mode.
- 5 In the Tracking Controls, click the panning control you want to automate: Left-Right (L-R), Front-Back (F-B), or Down-Up (D-U).
- 6 On the target track, click the Automation dropdown in the Track Header, then Pan > L/R Pan, Pan > F/B Pan, or Pan > U/D Pan to match your selection in the previous step.

You can increase the track height if the Automation dropdown is not visible.

- 7 Move the Playhead to the Timeline location where you want the automation to start.
- 8 Write a keyframe at the current position in the video by Option-clicking the Viewer at the location where you want to place the sound at that moment.

A corresponding Automation node is placed at the Playhead position on the automation line.

- 9 To add another keyframe, move the Playhead to another point in your timeline, Option-click the on-screen person or object you're tracking.

The Tracker Object jumps to that position, a keyframe is written, and a new Automation mode is added at the Playhead position.

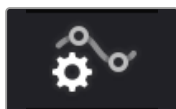
DaVinci Resolve will smoothly vector between the keyframes, as with any other automation curve data.

- 10 Continue creating keyframes at desired positions by repeating the previous step.
- 11 To create a "bounding keyframe" to lock a position that holds the automaton line before moving to a new position, Option-Shift-Click on a location.
- 12 Start playback to check, and if needed, adjust your automation.

Recording Mix Automation

Recording mix automation uses many of the same techniques as drawing but allows you to capture a performance where you are moving the controls you want to automate. This can be flexible and creative as you "perform to picture" to get the results you want.

To get started, you'll need to ensure that Automation recording is enabled (see "Enabling and Viewing Automation" above) and that one or more parameters are enabled for automation recording.



Clicking the Automation Controls button displays the Automation toolbar

Automation Controls

The Automation Controls button, to the right of the transport controls, lets you show and hide the Automation toolbar.

The Automation toolbar has buttons for each option that's available for preparing to record automation in your mix.



Automation toolbar options

The Automation toolbar displays the following options:

- **Automation Modes:** Controls how automation data is recorded.

Write: Records absolute changes to a control's position, replacing any data that was there previously.

Trim: Records relative changes to a control's position, maintaining relative existing levels created over time, but increasing or reducing those levels overall. This can be very useful when incrementally refining a mix.

- **Touch Modes:** Defines what happens when an automatable control is moved during a mix.

Off: No automation is recorded.

Latch: Automation is recorded once a control is moved and continues recording after the control is released until the transport is stopped.

Snap: Automation is recorded once a control is moved and then smoothly glides back to the pre-existing level at the point it was released, to create a seamless transition.

At the point where the glide transition ends, automation is no longer recorded unless the control is moved once again while the transport is playing. You can continue to capture additional automation with Snap until the transport is stopped.

The time for the glide transition can be set in milliseconds in Preferences > User > Fairlight > Automation. The default is 250 ms.

Snap Latch: A combination of Snap and Latch modes, where faders operate in Snap and all other controls operate in Latch. This can be useful when dynamically changing fader levels through a mix pass are desired, while other controls “stick” when they're moved. It is particularly useful when working with Fairlight mix control surfaces where faders and rotary controls can be changed simultaneously during a mix pass.

- **On Stop:** Defines what happens when entering Stop at the end of an automation pass.

Event: The last recorded automation value overwrites previous levels to the start of the next available recorded automation data (or event) in that track.

Hold: Deletes all previously recorded mixing data after what you've just recorded, to hold the last recorded level for the rest of that track.

Return: The last recorded automation value is interpolated to ramp back to the previously recorded automation values on that track.

- **Enables:** The following buttons let you enable or disable different controls for recording automation.

Fader: Automates track and bus volume.

Mute: Automates the mute button.

Pan: Automates all pan controls.

EQ: Automates all EQ controls.

Comp: Automates just the Compressor controls in the Dynamics window.

Gate: Automates just the Gate controls in the Dynamics window.

Lim: Automates just the Limiter controls in the Dynamics window.

Sends: Automates the Send controls in the Sends window.

Plugins: Automates all Fairlight, VST, or Audio Unit (AU) plugins.

Misc: Automates Buss on/off, Insert, and Direct Out controls. If a project is using the Fairlight legacy Fixed Bus configuration, it allows automation of fixed bus Main and Sub bus enables.

How to Record Automation

There are two different ways you can set up the recording of automation for levels, panning, EQ, Dynamics, and other audio controls in the mixer.

Recording Automation for Multiple Tracks

You can use the following steps to record automated changes to any audio control in any control strip:

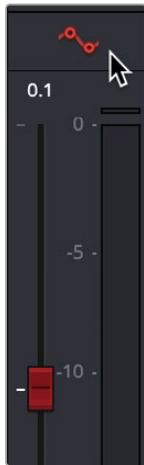
- 1 Open the Automation toolbar, and do the following three preparatory steps:
 - a) Choose whether you're going to write new automation, or trim automation that's already recorded. Write mode is appropriate when you're recording automation for the first time, or when you're overwriting previous automation with brand new values. Trim mode is appropriate when you're making incremental changes to previously recorded automation. Note that Trim mode works as a delta trim system only with fader positions at zero (unity) on Fairlight control surfaces at this time.
 - b) Next, choose Touch and On Stop behaviors that are appropriate for the kind of automation recording you need to do.
- 2 Move the playhead to the beginning of the section of the timeline you want to record automation for.
- 3 Next, initiate playback using any method (Spacebar, L, Play button, Fairlight or third-party audio control surface), and make whatever adjustments you want to the controls you've enabled automation recording for. As you make adjustments, the affected fader control turns red to let you know you're recording automation.
- 4 When you're finished, stop playback using any method (Spacebar, K, Stop button, third-party or Fairlight audio control panel). Automation recording stops as well.

Recording Automation for Specific Individual Tracks

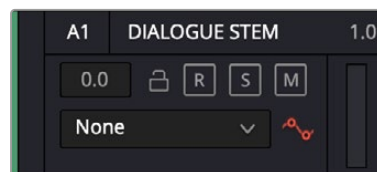
You can use the following steps to record automated changes to controls in specifically armed control strips:

- 1 Open the Automation toolbar, and do the following three preparatory steps:
 - a) Choose whether you're going to write new automation, or trim automation that's already recorded. Write mode is appropriate when you're recording automation for the first time, or when you're overwriting previous automation with brand new values. Trim mode is appropriate when you're making incremental changes to previously recorded automation.
 - b) Next, set Touch to Off, which disables across-the-board automation recording and requires you to arm specific tracks that you want to automate.

- 2 Click the automation arm button above the fader of any Mixer track or any track header to which you want to record automation. Even though the Touch control is set to off, moving a control on an armed channel strip will record automation in Latch mode.

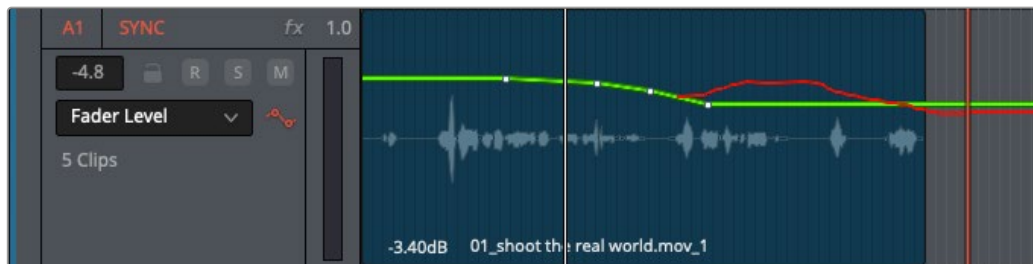


The automation arming button above a fader in the Mixer



The automation arming button in the track header

- 3 Move the playhead to the beginning of the section of the timeline you want to record automation for.
- 4 Next, initiate playback using any method (Spacebar, L, Play button, Fairlight or third-party audio control panel), and make whatever adjustments you want to the controls for which you've enabled automation recording. While you make adjustments, the affected fader control turns red to let you know you're recording automation. If you're displaying the same automation data in the Timeline that you're recording, you can see the new automation being drawn in real time, in red.



New automation line in red while being recorded

- 5 When you're finished, stop playback using any method (Spacebar, K, Stop button, Fairlight or third-party audio control panel). Automation recording stops as well. Displayed automation turns green once recording has stopped.

If you don't like what you've done, you can undo and start over, or you can edit the automation using methods described later in this chapter. Or, you can back the playhead up and overwrite automation at any time with new automation.

Automation Preview Mode

Preview is an additional mix automation workflow, specifically for working across scene-based material. When enabled, mix items that are in preview are not controlled by pre-recorded automation, so they respond manually to their controls. These can subsequently be placed into write (or trim) to write actual automation.

Typically, Preview mode is used to audition new mix settings for one particular section of a timeline, while other sections already have recorded automation data. Preview mode prevents pre-recorded automation from moving the controls you are trying to adjust in a targeted section. As soon as you're happy with the new adjustments, they can be written to the targeted section.

Preview mode frees the faders (and other controls) from automation control, and lets you move them freely while you experiment with different levels and settings. Ordinarily, moving one or more controls implies writing automation data for those controls, but entering Preview mode lets you play with the controls as much as you like without committing to anything, only writing automation data when you're ready.

To engage the Preview state on enabled mix items, first enter Preview mode by doing one of the following:

- Toggle Preview in the Automation toolbar.
- Press the Preview key on the Mix page of the Fairlight controller.

Once Preview is engaged:

- Individual parameters can be switched into Preview Touch Latch.
- You can use the AUTO key next to a fader to preview all enabled parameters on a channel.
- You can use the Auto button on the screen mixer strips.
- When you're in Preview mode, all parameters in Preview are indicated by a BLUE automation indicator.

Once in Preview mode, mix items can be placed into write (or trim) by:

- Dropping in manually via the Fairlight > Automation > Punch In menu choice.
- Dropping in manually with the In key on the Fairlight controller.
- Automatically using the Active In and Out points on the Fairlight controller.

Once enabled for Preview, parameters remain in that state regardless of transport starts and stops. This is different from putting mix items into WRITE, which must be done again after each transport stop.

Other Preview-related operations include:

- Filling a range defined by In and Out points with all parameters currently in Preview.
- Gliding all parameters from their existing values at the Range In point to the Preview values at the Range Out point.

Automation Active Range

The Automation Active Range allows you to set only a specific range on the timeline to record automation to, so that everything outside of that range is "write protected" and will not be disturbed.

- To enable the Active Range, choose Fairlight > Automation > Active Range, where you can mark the Active Range In/Out points, or enable or disable the Active Range.
- Once a range is defined, automation writing will only take place within the range.
- You can set the automation mode you wish to work in (e.g., Snap).
- Once playback begins, all automation data produced prior to the range automatically occurs in Preview mode (without having to enable the Preview switch), up and until you reach the Active Range in point. Once inside the Active range, automation recording will start.
- At the end of the range, automation recording is automatically punched out.
- The Active Range can be disabled by unchecking Fairlight > Automation > Active Range > Enable.

NOTE: Any controls in Read mode will play back their automation regardless of the status of the Active Range.



Active Range shown in the ruler with red bar; Automation data being written within the range

Viewing Automation in the Timeline

Once you've recorded automation for a supported plugin parameter or Mixer channel control, you can select and view it on the corresponding timeline track using one of the methods described below.

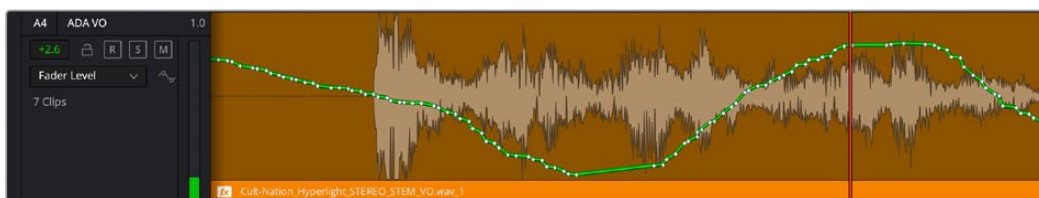
Right-Click to View Automation

Fairlight lets you select and view track automation by right-clicking parameters on Fairlight FX, supported Audio Unit or VST plugins, or Mixer channel controls. The parameter or control name is displayed in the track header, and the corresponding automation curve appears in the Timeline.

This time-saving method is especially useful when automating supported plugins with an extensive parameter list, such as a parametric EQ or mixer channel strip emulation.

The Automation Dropdown Menu

When automation is enabled via the Toggle Automation button, you can use the Automation dropdown menu on the track header to choose which automation curve to view, with options for Fader, Mute, Pan, EQ, Compressor, Limiter, Aux, Plugins, and Misc controls.

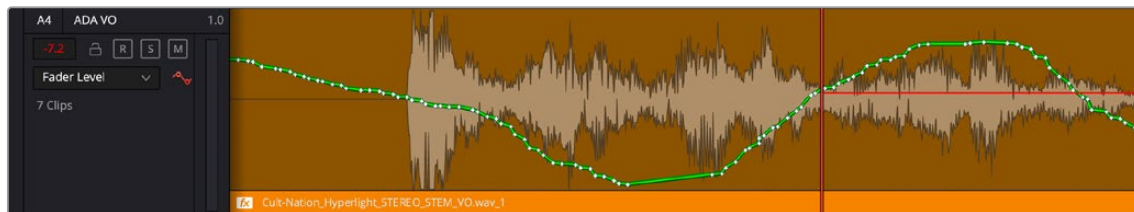


Mixing automation for the Fader shown in the Timeline

NOTE: When viewing Fader Level automation on a track that is assigned to a VCA, the VCA automation curve will also be visible.

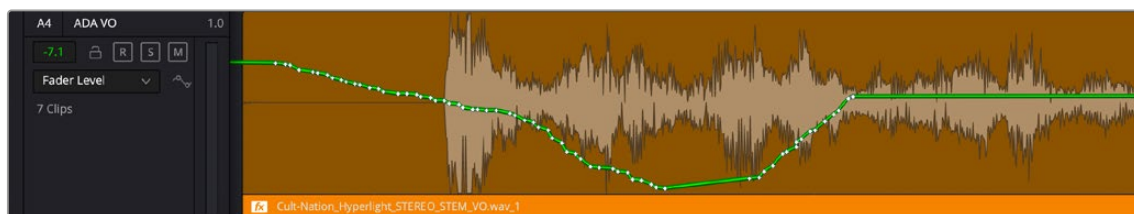
Overwriting Automation

Once you've recorded automation for a particular clip, you can overwrite that automation in subsequent passes. Using either of the previously described methods of recording automation, adjusting the levels of a track control with previously recorded automation displays a red line that shows the new level relative to the previously recorded level.



Setting new levels to overwrite previously recorded automation displays a red line

Pressing play to actually record this new automation will overwrite the previous levels at this new value.



New automation overwrites the previous levels

Automation data can be copied, pasted, and erased under the Fairlight Menu > Automation > Copy/Paste/Erase.

Automation Follows Edit

When editing your timeline, you can choose whether automation will follow the movement of clips or stay locked to its position. This is enabled by default but can be changed by unchecking Fairlight > Automation > Follows Edit.

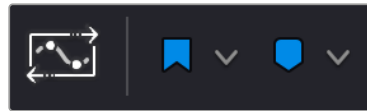
When enabled, anytime you move or edit a clip on the Timeline, the automation you'd written over its range will follow with it, ensuring that as you make changes to the edit, the automation for a given clip range will remain the same.

This is also extremely useful when doing an initial edit of audio clips that contain automation data. For instance, if you have a recurring sound effect that pans left to right, first automate that panning to the audio clip. When this clip is copied through the Timeline it will retain the same panning automation for each new instance. This can be very useful for a variety of edits that require the same automation data for audio clips in a timeline.

Be aware, however, that Automation Follows Edit will need to be turned off when copying and pasting clips or sequences where the automation that has been applied is not relevant to the new edits.

When Automation Follows Edit is enabled, automation data “lives with” the time range it came from. Any of that data, like volume or panning, will travel with clips. So if a complete section has been removed, all of the automation inside of the clips contained in that section are also removed. If a section of clips changes its location in the Timeline, all of the automation embedded in those clips move to the new location on the Timeline with the clips.

When automation is enabled in the Fairlight toolbar, the Automation Follows Edit button will appear.



The Automation Follows Edit button to the left of the Flag button on the toolbar

Editing Automation

When automation is enabled (Toggle Automation button is on), the Pencil tool appears. Combined with the Focus mode, these tools let you edit automation in different ways.



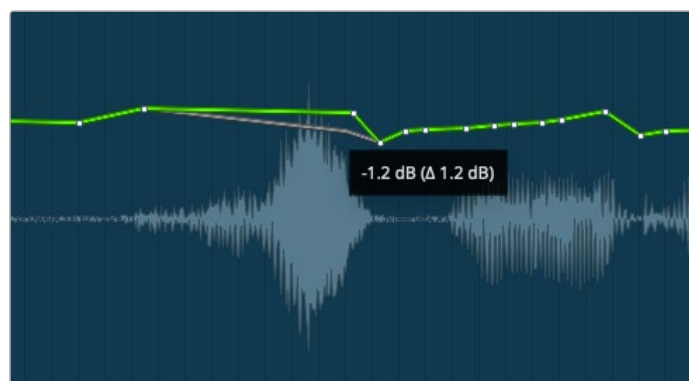
Focus mode provides the main tools used for editing automation. The Pencil appears when automation is enabled (Toggle Automation is on).

Adjusting and Deleting Automation Keyframes

The Focus tools let you adjust automation in two different ways.

Adjusting Individual Keyframes

Using the Focus tools, you can click and drag any automation keyframe up or down and left or right (bounded by its neighboring keyframes) to adjust it directly. When zoomed in, this can be an effective way of making precise adjustments. The segment of automation affected by the keyframe being dragged continues to show green to indicate the new curve being created, and a tooltip with the currently adjust value appears, while the previously created data is shown in gray. When you release the mouse button, the curve is adjusted



Single keyframe being adjusted showing prior level in gray

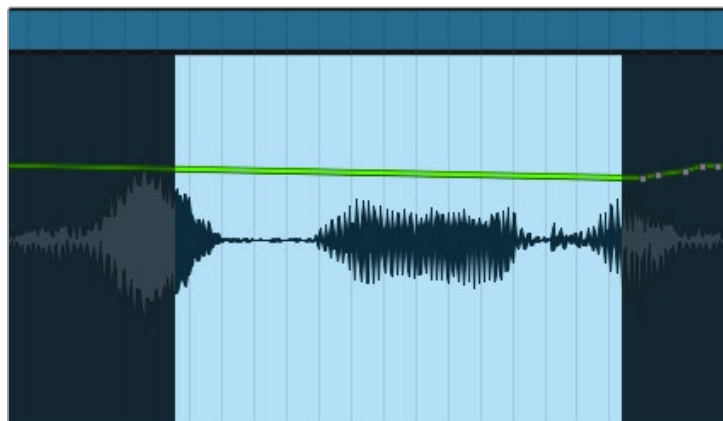
Adjusting and Deleting Multiple Keyframes

You can also use the Focus mode I-beam selector tool to create a selection of select multiple keyframes, or to move or delete them.



Use the Focus mode I-beam selector to select a range of keyframes

If you delete the keyframes in the range, the remaining section of the curve flattens out to fit the first and last keyframe of the selection.



Automation curve after deleting the selected keyframes

Adding and Editing New Keyframes

See “*Working with Audio Keyframes on the Timeline*” earlier in this chapter for more information about working with drawing and editing keyframes.

Clear All Track Automation

If you want to erase all automation on a track or bus, including all send and plugin automation, you can do so by right-clicking the track header and selecting Clear Track Automation from the menu. A new window will appear offering the following options for changing your mind or erasing the automation:

- **Cancel:** To keep your automation data.
- **Reset:** Clear all automation and return all affected controls and parameters to their default settings and values.
- **Hold:** Clear all automation and retain the settings and values of all affected controls and parameters.

Fairlight > Automation Controls

A series of commands in the Fairlight > Automation submenu let you initiate various automation tasks. These commands are:

- **All Read:** Switches all parameters that are in Trim or in Write mode back to Read mode.
- **Punch In:** When Preview mode is on, switches all parameters currently in Preview (blue) into Write or Trim. When Preview mode is off, this switches all Automation-enabled parameters on all selected channels into Write or Trim.
- **Punch Out:** When Preview mode is on, switches all parameters currently in Write, Trim or Preview into Read. When Preview mode is off, this switches all Automation-enabled parameters on all selected channels into Read.
- **Fill Range:** When there is an active In and Out range in the Timeline, the current value of all parameters in Preview will be written over that range.
- **Glide Range:** In Preview mode, “glides” all parameters from their existing values at the In point to the preview values at the Out point of the Timeline.
- **Join Mix:** Lets you manually resume writing automation in Latch mode from the current level and moving to any other.

Automation – Copy/Paste/Erase

The Fairlight > Automation submenu also has special local choices for Copy, Paste, and Erase of automation data. Copy/Paste allows you to do powerful range-based manipulation of automation, or to create a “snapshot” change where all automation data is pasted for a single point in time, allowing all enabled parameters to switch at once to new levels.

Using these commands requires that you choose which automation parameters are affected explicitly by using the Automation Enables controls (the same ones used to choose what automation data is recorded). For example, if you want to affect fader, mute, and pan automation, you must activate the enables for those parameters first. Only the parameters enabled are affected.

NOTE: Clip view or Automation view (any parameter) can be used with these commands; the choice of what is affected is governed by the Automaton Enables, not by the view.



Enables set for copy, paste, or erase of Fader, Mute, and Pan data only

- **Copy:** If there is an active In and Out range in the Timeline, Fairlight > Automation > Copy will copy the automation data of all selected channels and enabled parameters within this range to the clipboard. If there is no active In and Out range, this command copies the data of all automated parameters on all selected and enabled data at the playhead (Snapshot mode).
- **Paste:** Pastes the active range or in point for all automation data copied to the clipboard for Automation-enabled parameters.

If there is an active In to Out range defined, then the range will be filled with any automation data copied to the automation clipboard using Automation > Copy. If there is no active In to Out range defined, then the automation clipboard is simply pasted in its entirety.

TIP: Track Automation can be copied between Faders, Sends, and Direct-Out levels. For example, you can copy the Fader Level automation from one track to the Send Level of another, and vice versa.

The contents of the clipboard can potentially target a different channel set and/or with a different target time, just by using a new in point or track selection.

If there is an active In to Out range defined in the Timeline, and the clipboard was in Snapshot mode when the data was copied, then this range will be filled with static values of the snapshot data.

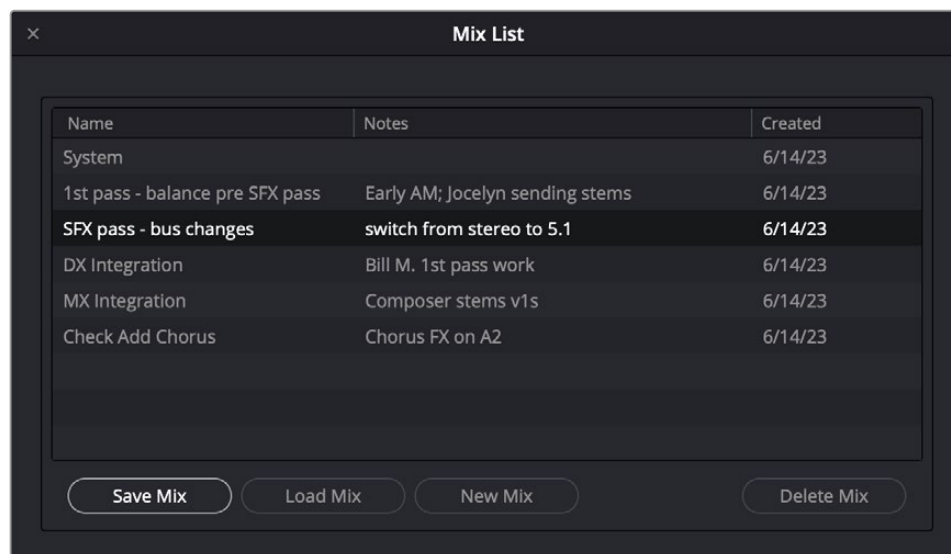
- **Erase:** If there is an active In and Out range in the Timeline, Erase will delete the automation data of all selected channels within the range, obeying all Automation Enable buttons. If there is no range selected, nothing is affected.

TIP: To clear all automation from a Timeline or group of tracks, select the entire timeline range, enable the Automation Enable buttons for the types of automation you wish to clear, and choose Fairlight > Automation > Erase. Alternatively, Fairlight > Automation > Mix List can be used to clear all automation from the Timeline.

Automation – Mix List

The Mix List allows you to create and manage multiple mix “passes” and complete routing and effects setups for a single Timeline. Each stored mix can have totally different mix automation data that is changing over time, routing, plugin setup, track grouping, etc.

This capability is incredibly powerful and allows a lot of creative freedom for creating different versions of mixes (for example moving between stereo, 5.1, and Dolby Atmos as formats), or between using totally different plugin effects or different creative approaches to the mix itself using all of the automated parameters.



The Mix List Window

So the Mix List's utility is not limited to storing mix automation data; it can be used to store a dynamic preset of literally anything about the setup of the Mixer.

Keep in mind that the Mix List is particular to a single Timeline and also cannot be shared between projects. However, using the Mix List can be more convenient than saving multiple different versions of timelines or projects.

The Mix List can act as a floating window where you can see the various mixes, or it can be hidden until you need to load or save a new version.

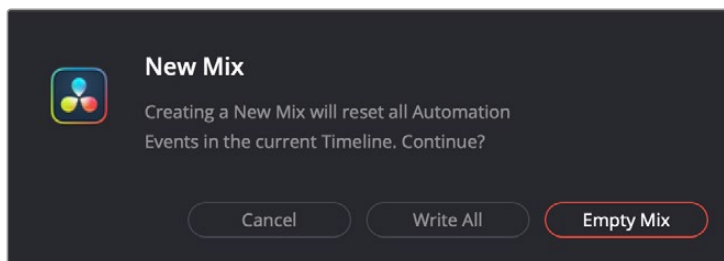
Using the Mix List Window

- Set up your mix, including all routing and automation as you like.
- When you're ready to save a mix version, choose Fairlight > Automation > Mix List, then choose "Save Mix" and give it a name, along with any notes you make want to add.
- You can then change your mix automation, any routing, etc. you wish and save a new version, load a different version, or delete an existing mix.
- The "System" preset is an internal system preset that can't be altered; it can be ignored.

The "New Mix" Dialog

For "New Mix" in the Mix List window, a dialog appears with the following options:

- **Cancel:** Lets you cancel the operation.
- **Write All:** Writes present levels of all controls to their automation curves establishing a baseline "snapshot" of all parameters.
- **Empty Mix:** Clears all automation data from the mix, leaving only the assignments, routing, track group enables, etc. This option offers a simple way to start "fresh" with a mixer setup on a new mix.



New Mix dialog

Playing Automation

After you've automated a mix, playing it back is as simple as moving the playhead to an area of the timeline prior to the recorded automation, and initiating playback. As the timeline plays, the onscreen controls for each automated parameter are shown in green. They move and update to show their recorded levels.



Controls that have recorded automation during playback are shown in green.